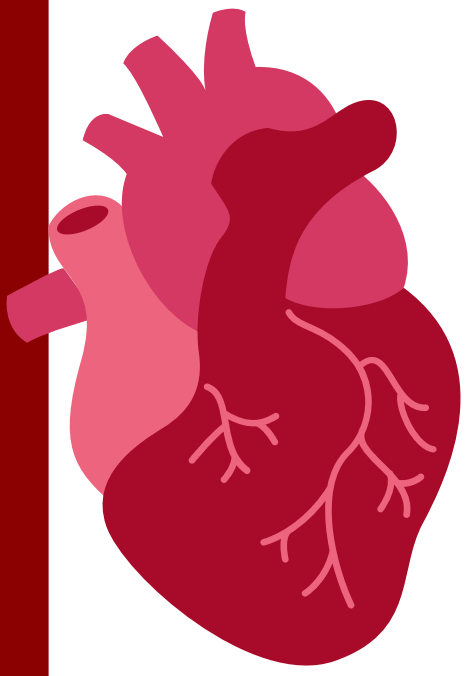
سُبْحان الله وبحمده، سُبْحان الله العظيم

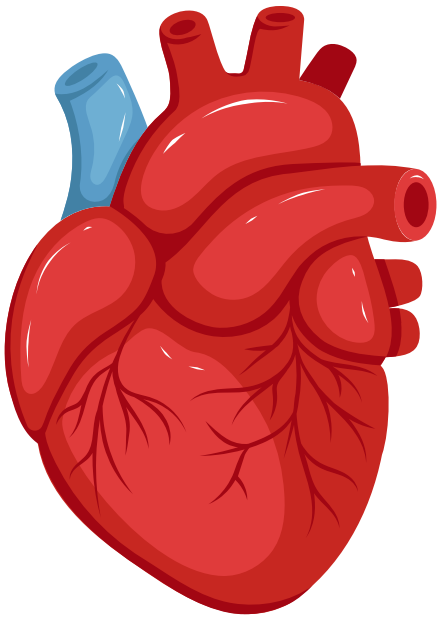


MED25

Circulation & Breathing Module Testbank

CVS Previous Years

اللهم علّمني ما ينفعني، وانفعني بما علّمتني، وزدني علمًا.



Second Mid

Lessons Included:

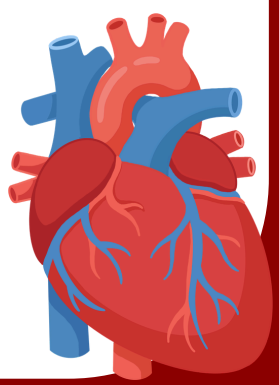
1. **Vessels and vascular tumors**
2. **Vasculitis**
3. **Pathology of Hypertension**
4. **Pathology of valvular Heart Diseases**
5. **Endocarditis**
6. **Development of the heart and its great vessels**
7. **Congenital anomalies and abnormal development of heart and its great vessels**
8. **Antihypertensive drugs 1**
9. **Antihypertensive drugs 2**
10. **Antiarrhythmic drugs**
11. **Physiology of vascular endothelium**
12. **Cardiac cycle**
13. **Normal ECG**
14. **Abnormal ECG and cardiac arrhythmias**

UFC QS med19 midterm _protected

10) A primitive structure that develops into auricle of left atrium?
A- Sinus venosus
B- Pulmonary vein
C- Bulbos cortus
D- Primitive atrium

9) Which of the following will develop heart?
A- visceral (spalanic) layer
B- soma=c mesoderm
C-ectoderm
D- endoderm

10	**9**	**#**	**#**
D	**A**		



Final 2

Embryo- normal development of the heart

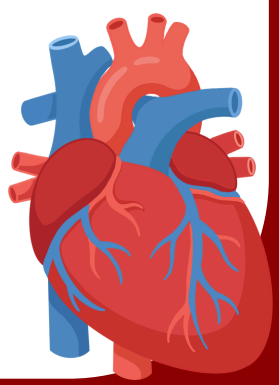
1- The rough part of the left ventricle develops from:
A. Primitive ventricle
B. Proximal 1/3 of bulbus cordis
C. Sinus venosus
D. Conus cordi

2- The vascular system develops at:
A. 2nd week
B. 3rd week
C. 4th week
D. 5th week

3-Which of the following primitive vessels contributed to the development of the heart?
A. Left common cardinal
B. Right common cardinal
C. Left umbilical
D. Right umbilical

4-Which of the following helps with the development of the left auricle?
A. Pulmonary veins
B. Primitive atrium
C. Sinus venosus
D. Bulbus cordis

1	2	3	4
A	B	B	B



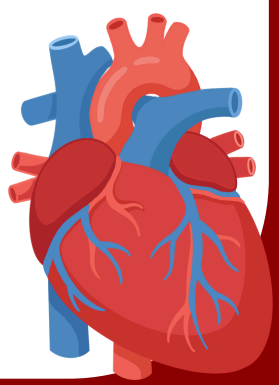
Final 2

5-Which of the following is the primitive structure of the coronary sinus?
A. Left sinus horn of sinus venosus
B. Right sinus horn of sinus venosus
C. Right vitelline vein
D. Right common cardinal vein

7-A newborn has a defective coronary sinus, and the cause is most likely congenital. Which part of the primitive heart tube is most likely responsible for the defect?
A. Sinus venosus
B. Primitive atrium
C. Primitive ventricle
D. Bulbus cordis

8- Which of the following primitive structures develops into the left auricle?
A. Primitive atrium
B. Bulbus cordis
C. Pulmonary veins
D. Sinus venosus

9- The vestibule of the left ventricle is formed from?
A. Sinus venosus
B. Primitive atrium
C. Primitive ventricle
D. Bulbus cordis



5	7	8	9
A	A	A	D

Final 2

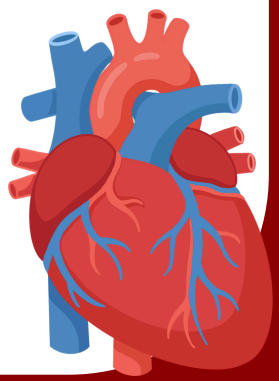
10-A patient has a thin outflow part of the right atrium. What is the most likely defect?
A. Body & right sinus horn
B. Conus cordis
C. Primitive ventricle
D. Sinus venosus

Embryo - abnormal development of the heart

1-After birth, the patent ductus arteriosis is an abdominal connection between:
A.Arch of aorta and pulmonary trunk
B.Right and left atrium
C.Right and left ventricle
D.Ascending aorta and pulmonary trunk

2-ostium secundum is a defect in:
A) Upper part of septum primum
B) Lower part of septum primuim
C) Upper part of septum secundum
D) Lower part of septum secundum
(مو مذكورة نصا بس بالفهم تنحل)

3-with potent ductus arteriosis the blood :
A.Moves from pulmonary trunk to aortic arch
B.Moves from pulmonary trunk to ascending aortic
C.Moves from umbilical vein to SVC
D.Moves from umbilical artery to placenta



10	1	2	3
A	A	A	A

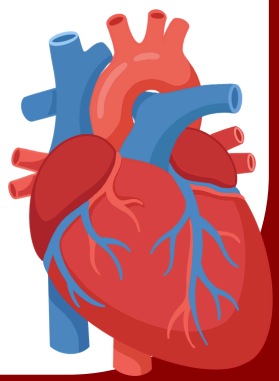
Final 2

4-which of the following is not included in the tetralogy of Fallot?
A.Overriding aorta
B.Hypertrophy of right ventricle
C.Atrial septal defect
D.Pulmonary stenosis

5- A patient was found to have their aorta originating from the right ventricle and the pulmonary arteries originating from the left. What is the name of this defect?
A. Defect in the interatrial septa
B.Transposition of the great vessels
C.Tetralogy of Fallot
D. Persistent truncus arteriosus

6-defect in the heart that cause overrides both ventricles and receives blood from both sides what is the reason?
A.Interventricular septal fail to develop
B.Conotruncal septal fail to develop
C.Valvular defect
D.open ductus arteriosus

7-what structure can cause Atrial septal defects (ASD) through the opening of?
A.Foramen ovale
B.Patent ductus venosus
C.Patent ductus arteriosus
D.Persistence of truncus arteriosus.



4	5	6	7
C	B	B	A

Final 2

8-Which of the following embryonic structures allows fetal blood to circulate from the pulmonary trunk to the descending aorta before birth?

A. Ostium primum
B. Ostium secundum
C. Ductus arteriosus
D. Ductus venosus

9-A child presents to the ER, and after a series of procedures, you determine that his aorta is on the right side of his pulmonary trunk. What is the cause behind this defect?

A. Failure of conotruncal septum to follow its spiral course
B. Unequal division of the conus
C. Failure of conotruncal ridges to fuse
D. Atrial septal defect

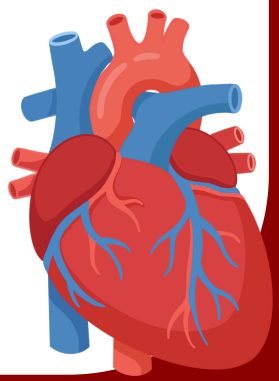
10-Pulmonary stenosis, VSD, overriding aorta, and ventricular hypertrophy are characteristic of which of the following?

A. Tetralogy of Fallot
B. Transposition of great arteries
C. Atrial septal defect
D. Patent ductus arteriosus

11- A patient has ventricles that share one outflow tract. What is the most likely defect?

A. Ventricular septal defect
B. Conotruncal septum not formed
C. Incomplete development of endocardial cushions
D. Persistent truncus arteriosus

8	9	10	11
C	A	A	D



Final 2

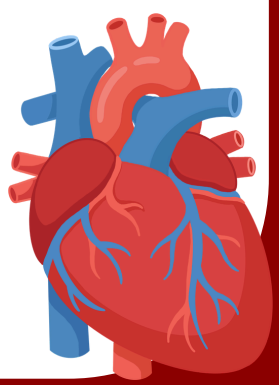
12-Failure of the conotruncal ridge to twist results in which condition?
A. Transposition of the great vessels
B. Ventricular septal defect
C. Atrial septal defect
D. Persistent truncus arteriosus

13-Which embryonic structure closes to form the ligamentum teres?
A. Umbilical vein
B. Distal part of umbilical arteries
C. Proximal part of umbilical arteries
D. Vitelline vein

SDL anatomy-radiology

1. The imaging technique used to follow up a patient with aortic valve stenosis is:
A. Echocardiograph
B. MRI
C. Angiography
D. X-ray of skull

2. A newly born baby is born cyanotic and having difficulty breathing. On examination, a well-heard diastolic murmur suggests valvular stenosis. The imaging technique used is:
A. Echocardiogram
B. MRI
C. Angiography
D. X-ray of the skull



12	13	1	2
A	A	A	A

Final 2

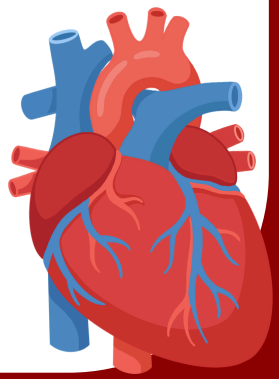
L8- Physiology,Normal ECG

1- What view of the heart does lead II, III, and , AVF represent?
A. Inferior
B. Anterior
C. Lateral
D. Septal

2- Lead 2 is :
A. +ve in Right arm and -ve in left arm
B. +ve in Right arm and -ve in left leg
C. +ve in Left leg and -ve in Right arm
D. +ve in Left leg and -ve in Left arm

3- What wave in the ECG represents repolarization of both ventricles?
A. T wave
B. QRS Complex
C. P wave
D. P-R interval

4- Which of the following is correct regarding the electrical impulse that generates
from the SA node as seen on an ECG?
A. QRS complex (0.8 second)
B. PR interval (0.12-0.2 seconds)
C. T wave (0.25 second)
D. QT interval (0.3-0.4 seconds)



1	2	3	4
A	C	A	B

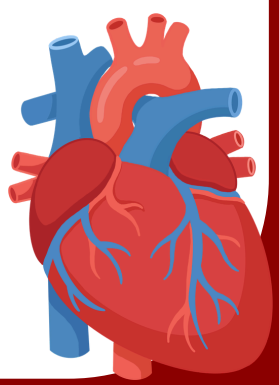
Final 2

5- What wave in the ECG represents ventricular repolarization?
A. Q wave
B. R wave
C. T wave
D. P wave

6- The ventricle becomes completely depolarized in ECG accompanied with the plateau in which of the following phases ?
A. QRS complex
B. QT interval
C. ST segment
D. T wave

7- Ventricular repolarization in ECG is represented by?
A. QRS complex
B. P wave
C. T wave
D. U wave

8- In a healthy heart, the QRS waves correspond to?
A. Ventricular repolarization
B. Ventricular depolarization
C. Atrial repolarization
D. Atrial depolarization



5	6	7	8
C	C	C	B

Final 2

9- A patient has an elevated ST segment and inverted T wave. What is the likely diagnosis?

A. Acute MI (ischemic heart disease)
B. Atrial fibrillation
C. Myocardial ischemia
D. Pulmonary embolism

10- What indicates QRS in ECG?

A. ventricular depolarization
B. ventricular repolarization
C. plateau
D. Atrial depolarization

11- ST elevated in the ECG indicates that the patient has ?

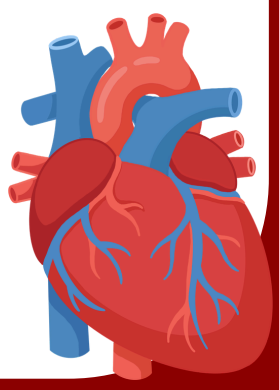
A. Myocardial infarction
B. Myocardial ischemia
C. Tachycardia
D. Atrial Fibrillation

12- What does the ECG show ?

A. The ECG shows elevated S-T segment
B. The ECG shows elevated T wave
C. The ECG shows prolonged P-R interval
D. The ESG shows inverted S-T segment



9	10	11	12
A	A	A	A



Final 2

L9 - Physiology, Cardiac Arrhythmias

1- A patient was given an ACE inhibitor and developed arrhythmia. What is the most likely ion imbalance causing this, given that a tall, peaked T-wave is visible on the ECG?

A. Hyperkalemia
B. Hypokalemia
C. Hypercalcemia
D. Hypocalcemia

2- In first-degree heart block what can be expected ?

A. PR interval is less than 0.12 (3 small squares)
B. PR interval is prolonged
C. P wave is sometimes followed by QRS
D. P wave is not followed by QRS

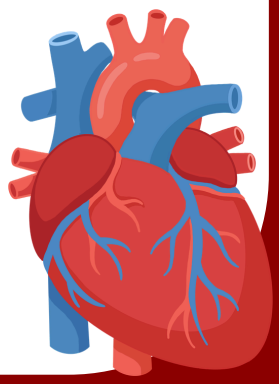
3- What make Narrow and tall peaked T wave in ECG?

A. Hyperkalemia
B. Hypokalemia
C. Hypocalcemia
D. Hypercalcemia

4- A patient is noted to have an abnormally shortened PR interval on their ECG. What could be the cause?

A. Atrial fibrillation
B. Right bundle branch block (RBBB)
C. Left bundle branch block (LBBB)
D. Wolff-Parkinson-White (WPW) syndrome

1	2	3	4
A	B	A	D



Final 2

5- What is the cause of the ecg in the picture above?

A. Hyperkalemia
B. Hypokalemia
C. Hypercalcemia
D. Hypocalcemia



6- An ECG shows no P-waves and an irregular rhythm. What does this represent?

A. Atrial fibrillation
B. Sinus tachycardia
C. Ventricular extrasystole
D. Ischemia

7- In ECG tall Peaked T-wave with "tent" like appearance indicates?

A. Hyperkalemia
B. Hypokalemia
C. Hypercalcemia
D. Hypocalcemia

8- Which ECG shows ventricular extrasystole?

A.



B.



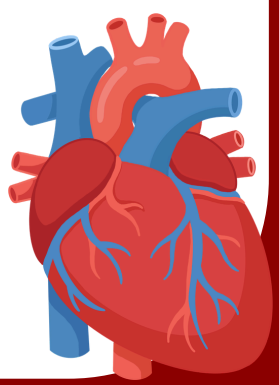
C.



D.



5	6	7	8
A	A	A	C



Final 2

Physiology: Cardiac cycle

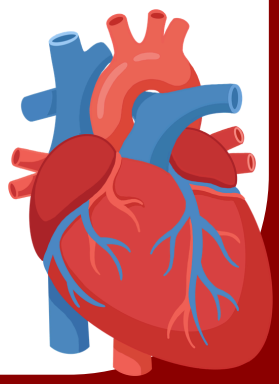
1.Which one of the pairing statements is correct?
A. First heart sound - closure of semilunar valves
B. 3rd heart sound - elderly people
C. Isovolumetric relaxation phase - sharp decrease in ventricular pressure
D. Right Atrial systole - passive filling of the ventricle

2.What will happen at the beginning of the isovolumetric contraction?
A. Aortic pressure increased
B. Aorta expansion
C. Ventricle volume decreased
D. A-V valves closure

3.Regarding Ventricular Systole?
A. Depends on intracellular Ca++
B. Ends by the end of plateau
C. Corresponds P wave in the ECG
D. It occurs after atrial diastole

4.When the ventricular pressure is lower than the atrial pressure this leads to:
A. AV valve open
B. Aortic outflow of the blood
C. Right ventricular contraction
D. Semilunar valve open

1	2	3	4
C	D	B	A



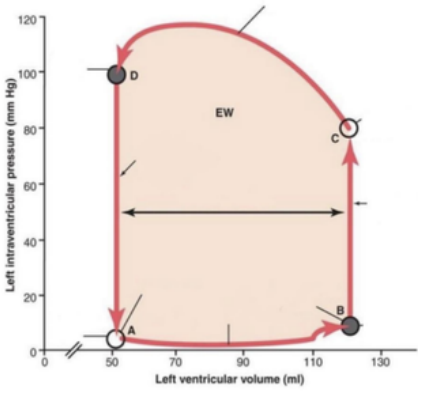
Final 2

5.Between AV valves closure and opening of Semilunar valves is the following phase:
A. Isometric relaxation
B. Rapid ejection
C. Isometric contraction
D. Reduced ejection

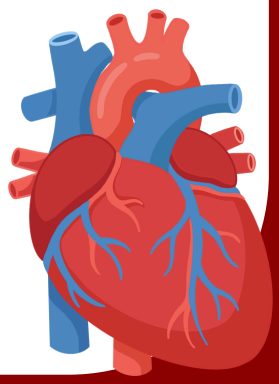
6. The lowest ventricular volume occurs during which phase?
A. Isovolumetric ventricular relaxation
B. Isovolumetric ventricular contraction
C. Rapid ventricular ejection
D. Reduced ventricular ejection

7. QRS complex of ECG occurs immediately before which phase?
A. Isometric contraction
B. Isometric relaxation
C. Atrial systole
D. Maximum filling phase

8. Which line represents the Isovolumetric relaxation phase ?
A. Line A-B
B. Line B-C
C. Line C-D
D. Line D-A

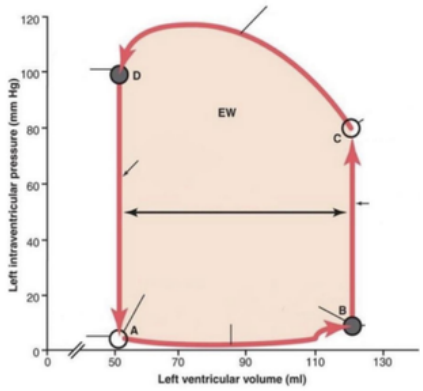


5	6	7	8
C	A	A	D



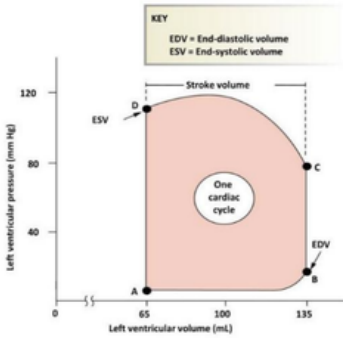
Final 2

9. At which point
the closing of the mitral valve will occur ?



A. A
B. B
C. C
D. D

10. Which line represents the isovolumetric relaxation phase in the pressure-volume loop of
the left ventricle?



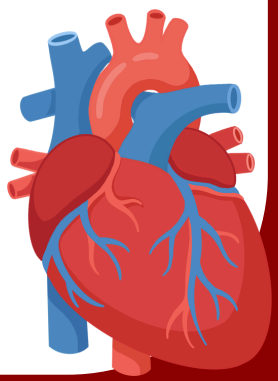
A. Line D-A
B. Line B-C
C. Line C-D
D. Line A-B

11. In the ejection phase, the atrioventricular valves ……,and the semilunar valves…… .

A. Close, open
B. Close, close
C. Open, close
D. Open, open

12. The phase of the cardiac cycle between mitral valve closure and aortic valve opening?

A. Isovolumetric contraction phase
B. Atrial systole
C. Ventricular systole
D. Isovolumetric relaxation phase

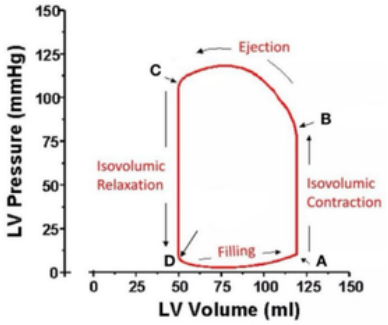


9	10	11	12
B	A	A	A

Final 2

13. Point A represents?

A. Mitral valve closure
B. Mitral valve opening
C. Aortic valve opening
D. Aortic valve closure

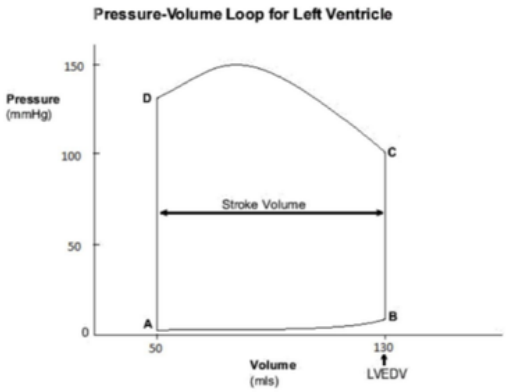


14. On cardiac examination, the patient's rhythm was normal with S4 at the apex.
What characterizes the S4 sound?

A. It is caused by atrial relaxation during ventricular ejection
B. It can be heard in patients with left ventricular hypertrophy
C. It can be heard with stethoscope in the diaphragm
D. It has a high frequency, usually 600 cycles/sec

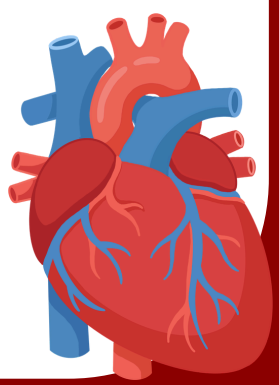
15. At which point does the mitral valve closure occur?

A. B
B. A
C. D
D. C



16. The graph shows ventricle pressure-volume relationship; point A is:

ANSWER: Mitral valve closure



13	14	15	16
A	B	A	-

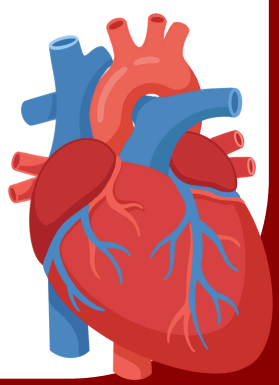
Final 2

17. Reduced ejection fraction occurs in which of the following ?
A. Systolic dysfunction
B. Diastolic dysfunction
C. Exercise
D. Normal regulation of blood pressure

18. If there was an infarction of the AV node that caused an abnormal increase in the AV nodal delay, what will this affect?
A. Ventricular filling
B. Isovolumeric relaxation
C. Atrial systole
D. Ventricular systole

20. Arteries are elastic and recoil during diastole, why?
A. To maintain blood pressure
B. To maintain constant blood flow
C. To maintain blood volume
D. For exchange

17	18	20
A	A	B



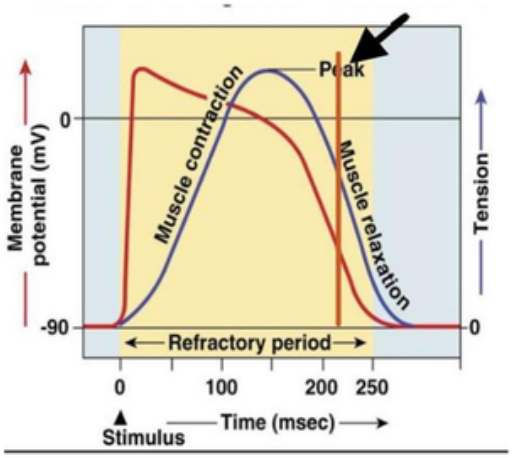
MED20 midterm

2) Which of the following is not a classic ECG change in MI:

a. Double P wave
b. Inverted T wave
c. ST elevation
d. Abnormal development of deep Q wave

5) The red line in the figure separate ARP from RRP which statement is correct?

a. RRP is occurring during depolarisation
b. ARP in cardiac muscle is longer than skeletal muscle
c. During ARP can case premature impulse

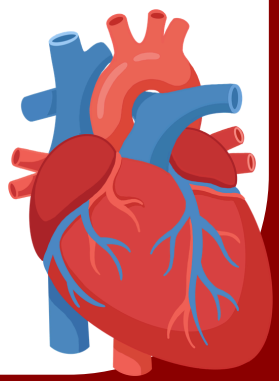


6) What wave in the ECG represents repolarization?

a. T-Wave
b. QRS Complex

7) Resistance to blood flow?

a. Arterioles
b. Aorta
c. Veins
d. Capillaries



2	5	6	7
a	b	a	a

MED20 midterm

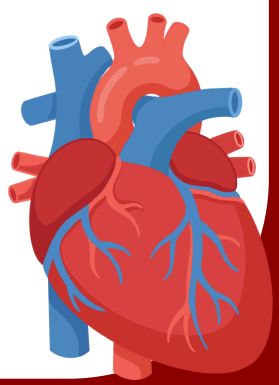
9) Which of the following is correct regarding to the electric impulse that generates from the SA node in ECG?
a. QRS complex (0.14-0.8)
b. PR interval (0.12-0.2)

10) What increases the force of cardiac contractility?
a. Ca2+ channel blocker
b. Muscarinic blocker
c. Digitalis

12) A patient is noted to have an abnormally shortened PR interval on their ECG, What could he have?
a. Right bundle
b. Left bundle
c. WFW

15)Which of the following is NOT a classic change in ECG in myocardial infarction?
a. Inverted T wave
b. ST segment elevation
c. Development of an abnormal Q wave
d. Double P wave

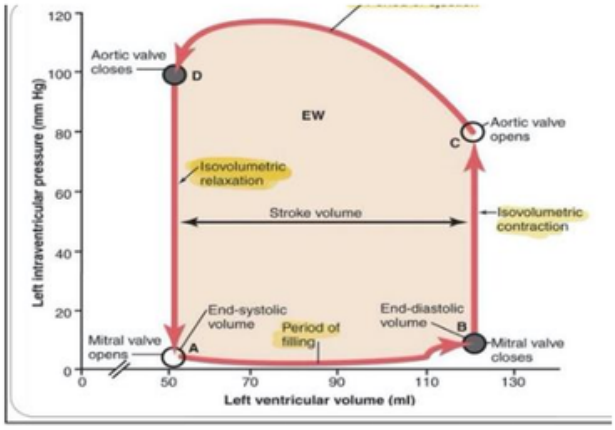
9	10	12	15
b	c	c	d



MED20 midterm

16) Which line represents the isovolumetric relaxation phase?

a. Line A-B
b. Line B-C
c. Line C-D
d. Line D-A



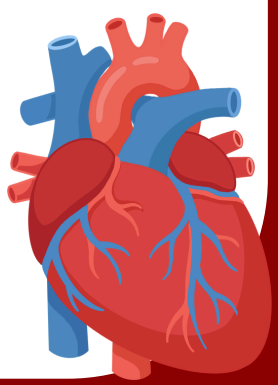
21) A patient was found to have their aorta originating from the right ventricle and the pulmonary arteries originating from the left. What is the name of this defect?

a. Defect in the interatrial septa
b. Transposition of the great vessels

23) Inflammation is the current favorites hypothesis for atherosclerosis. Which of the following diagnostic marker would be indicative of increased inflammation?

a. HDL
b. Hypercysteinemia
c. C-Reactive protein
d. CK

16	21	23
d	b	c



MED20 midterm

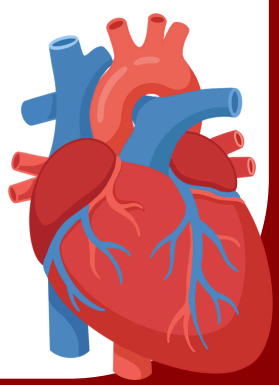
19) What disease is LDL a risk factor for?
a. Sickle cell anemia
b. Liver cancer
c. Intestinal Obstruction
d. Chronic heart disease

24) case A man presents with aortic aneurysm and a palpable abdominal mass what's the pathogenesis ?
a. Cyanotic degeneration of media
b. Fibrofatty plaques in intima

26) Which of the following primitive vessels shared in the development of the heart?
a. Left common cardinal
b. Right common cardinal
c. Left umbilical
d. Right umbilical

31) Blood pressure is described as:
a. Cardiac output times peripheral resistance
b. Stroke volume times heart rate divided by total peripheral resistance
c. Stroke volume times heart rate
d. Cardiac output divided by stroke volume

19	24	26	31
d	b	b	a



MED20 midterm

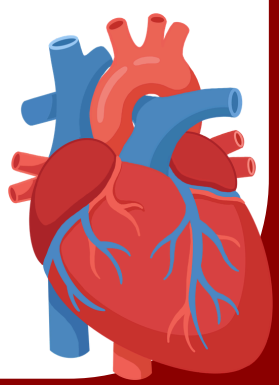
32) Which of the following helps with development of the left auricle?
a. Pulmonary veins
b. Primitive atrium

33) AV nodal cells have?
a. The lowest firing potential in the cardiac conduction system
b. Frequent gap junctions
c. Fast conduction velocity
d. Unstable membrane potential

34) Total peripheral resistance (TPR) mainly depends on?
a. Diameter of capillaries
b. Diameter of arterioles
c. Velocity of blood
d. Length of blood vessel

8) Blood pressure increase and heart rate decrease due response to:
a. Hemorrhage
b. Exercise
c. Increased Intracranial Pressure

32	33	34	8
b	d	b	c



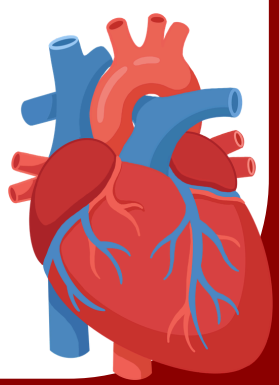
CVS MID-EXAM 24

Q13. Which of the following is a vasoconstrictor?
A. Endothelin-1
B. Nitric oxide
C. Prostacyclin
D. EDHF

Q14. Which of the following is associated with primary (essential) hypertension?
A. Increased peripheral resistance
B. Renal disease
C. Neurogenic tumour
D. Endocrine disease

Q17. A patient with hypertension was stable on a drug but later required another drug that caused fatigue and sexual dysfunction. The new drug is most likely:
A. Metoprolol
B. Captopril
C. Losartan
D. Amlodipine

Q26. A woman has a family history of hypertension, consumes a high-salt diet, and has a sedentary lifestyle. Her hypertension does not improve by drugs alone. Why?
A. Environmental factors amplify her genetic susceptibility
B. Her blood volume is always decreased
C. Her sympathetic activity is completely absent
D. Her kidneys cannot produce renin



13	14	17	26
A	A	A	A

CVS MID-EXAM 24

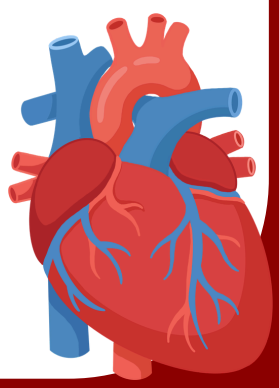
Q36. What does the QRS complex represent on an ECG?
A. Ventricular depolarization
B. Ventricular repolarization
C. Atrial repolarization
D. Atrial depolarization

Q38. A patient with heart failure and hypertension is on hydralazine. Which drug combination is most appropriate to control the condition?
A. Digoxin + Diuretic
B. β-blocker + Hydrochlorothiazide
C. ACE inhibitor + Digoxin
D. Calcium-channel blocker + Nitrate

Q23. A patient has regurgitation due to damage to a structure of the heart valve. Which structure is most likely affected?
A. Papillary muscle
B. Chordae tendineae
C. Trabeculae carneae
D. Moderator band

Q12. Which of the following are common primary sites of cancer that can metastasize to the heart?
A. Lung and breast
B. Thyroid and skin
C. Kidney and pancreas
D. Brain and bone

36	38	23	**12**
A	**B**	**B**	**A**



CVS MID-EXAM 24

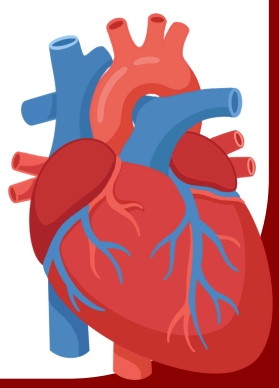
Q17. A patient with hypertension was stable on a drug but later required another drug that caused fatigue and sexual dysfunction. The new drug is most likely:
A. Metoprolol
B. Captopril
C. Losartan
D. Amlodipine

Q18. A 50-year-old man has controlled hypertension. An ECG shows a premature ventricular complex. Which of the following is the diagnosis?
A. Premature ventricular complex (PVC)
B. Premature atrial contraction
C. First-degree AV block
D. Sinus bradycardia

Q19. A short QT interval is associated with:
A. Hypocalcemia
B. Hypercalcemia
C. Hyperkalemia
D. Hypokalemia

Q29. A patient presents with nodules on his feet. Which infectious agent is associated with this condition?
A. Human herpesvirus-8 (HHV-8)
B. Epstein–Barr virus
C. Cytomegalovirus
D. Hepatitis B virus

17	18	19	29
A	A	B	A



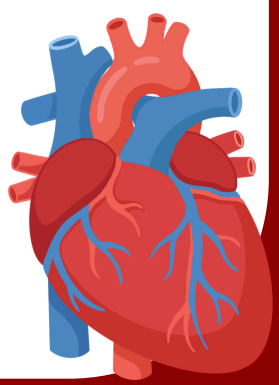
Vit.CVS Midblock_protected

1- After birth, the patent ductus arteriosis is an abdominal connection between:
A) Arch of aorta and pulmonary trunk
B) Right and left atrium
C) Right and left ventricle
D) Ascending aorta and pulmonary trunk

13- ostium secundum is a defect in:
A) Upper part of septum primum
B) Lower part of septum primuim
C) Upper part of septum secundum
D) Lower part of septum secundum

14- Which one of the pairing statements is correct?
A) first heart sound - closure of semilunar valves
B) 3rd heart sound - elderly people
C) iso-volumetric relaxation phase - sharp decrease in ventricular pressure
D) Right Atrial systole - passive filling of the ventricle

15- hypertension can be age related. A rising in systolic pressure can be considered a part of normal aging process. Why does systolic pressure increase?
A) Auto-regulation
B) decrease in total peripheral pressure
C) excess work load on heart
D) decrease elasticity of large artery



1	13	14	15
A	A	C	D

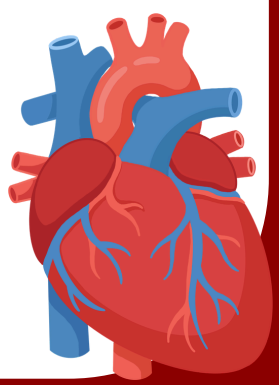
Vit.CVS Midblock_protected

17- The rough part of the left ventricle develops from:
A) primitive ventricle
B) proximal 1/3 of bulbus cordis
C) sinus venosus
D) cones cordis

22- What will happen at the beginning of the isovolumetric contraction?
A) aortic pressure increased
B) aorta expansion
C) Ventricle volume decreased
D) A-V valves closure

26- When the ventricle pressure is lower than in atrial pressure this leads to:
A) AV valve open
B) aortic outflow of the blood
C) right ventricular contraction
D) semilunar valve open

29- the mechanism of action of digitalis :
A) block Ca and Na exchanger
B) block Na k+ pump
C) ca efflux
D) increases The slope of SA node
E) Decreased calcium inflow
F) Stimulate beta-1 receptor



17	22	26	29
A	D	A	B

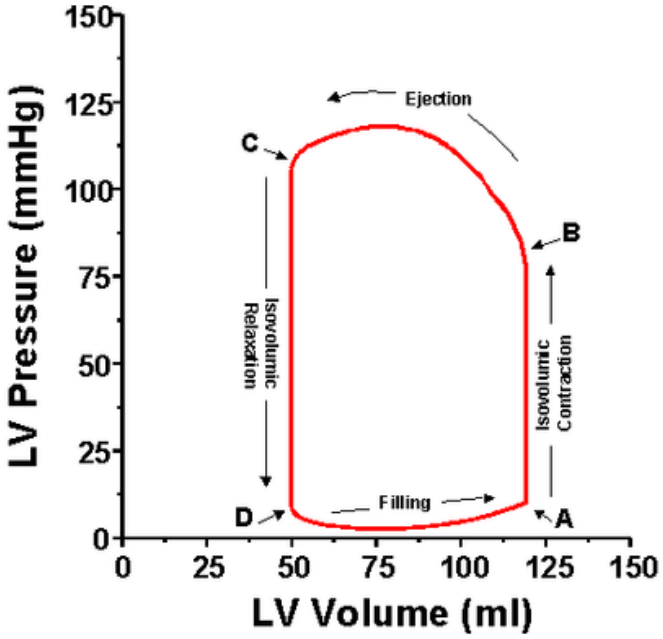
Vit.CVS Midblock_protected

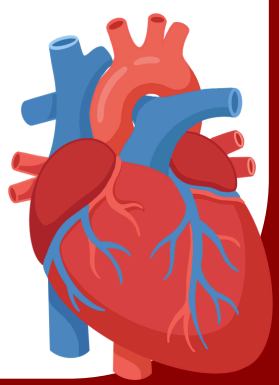
32- with potent ductus arteriosis the blood:

A) moves from pulmonary trunk to aortic arch
B) moves from pulmonary trunk to ascending aortic
C) moves from umbilical vein to SVC
D) moves from umbilical artery to placenta

35- Point A represents?

A) Mitral valve closure
B) Mitral valve opening





32	35
A	A

Vit.CVS Final_protected

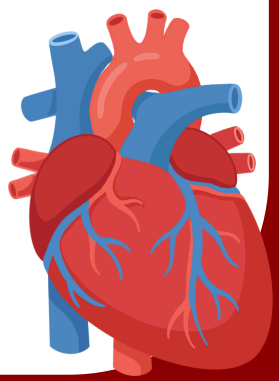
5. Decrease force of ventricular contraction occurs in
A) vagal stimulation
B) b1 agonist
C) Calcium channel blockers
D) Digitalis

15. What view of the heart does lead II, III, and , AVF represent?
A) inferior
B) Anterior
C) Lateral
D) Septal

18. ECG with no P-waves and irregular rhythm represents:
A) Atrial fibrillation
B) sinus tachycardia
C) Ventricular extrasystole
D) Ischemia

22. What drug is contraindicated in a hypertensive patient with arterial renal stenosis?
A) Lisinoprel
B) Clonidine
C) Hydralazine
D) Diltiazem

25. which of the following is not included in the tetralogy of Fallot?
A) Overriding aorta
B) Hypertrophy of right ventricle
C) Atrial septal defect
D) Pulmonary stenosis



5	15	18	22	25
C	A	A	A	C

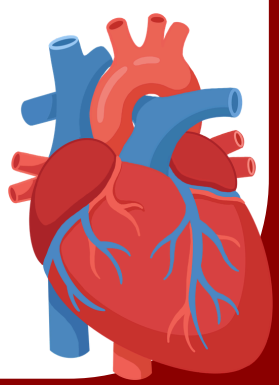
Vit.CVS Final_protected

26. Patient with blood pressure of 240/130 what is the expected microscopic change in the arterioles?
A) Deposition of homogeneous plasma protein material
B) Concentric proliferation of smooth muscle cell

27. In ECG tall Peaked T-wave with "tent" like appearance indicates:
A) Hyperkalemia
B) Hypokalemia
C) Hypercalcemia
D) Hypocalcemia

29. 12 years old boy developed beta-hemolytic streptococcus pharyngitis which resolved spontaneously, however three weeks later he presented with a skin rush and his doctor hear murmur on physical examination, which of the following changes is least likely to be found in the heart of this boy at this stage of his disease?
A) Aschoff nodules in the myocardium
B) Fibrosis of the mitral valve with fusion of the commissures
C) Foci of fibrinoid necrosis in mitral valve
D) Fibrinous pericarditis

34. What is the nature of anitschow cells ?
A) Activated macrophage
B) T lymphocyte
C) Plump neutrophils
D) mast cells



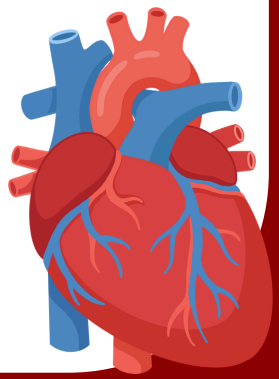
26	27	29	34
B	A	B	A

Vit.CVS Final_protected

36. year old woman ... blood cultures are positive for gram positive bacteria. Urine analysis show hematuria. An ecg shows a 1.5 cm vegetation on the mitral valve. Which of the following conditions is this patient most likely to develop?
A) cerebral arterial mycotic aneurysm
B) dilated cardiomyopathy
C) myxomatous mitral valve degeneration
D) pulmonary abscess

37. At birth, a patent ductus arteriosus is an abnormal connection between:
A) left & right atria
B) pulmonary trunk & arch of aorta
C) left & right ventricle
D) pulmonary trunk and ascending aorta

39. A 71-year-old woman has had a 10-kg weight loss accompanied by severe nausea and vomiting of blood for the past 8 months. On physical examination, she is afebrile. Biopsy specimens obtained by upper gastrointestinal endoscopy show adenocarcinoma of the stomach. Which of the following cardiac lesions is most likely to be present in this woman?
A) Calcific aortic valvular stenosis
B) Constrictive pericarditis
C) Left ventricular mural thrombosis
D) Nonbacterial thrombotic endocarditis



36	37	39
A	B	D

Vit.CVS Final_protected

42. 30 years old patient known that he have HIV infection for several years. Complained of rapidly spread Multiple red-purple skin plaques and nodules. Whats the virus associated with this case?
A) human herps virus 8
B) B. human pappiloma virus

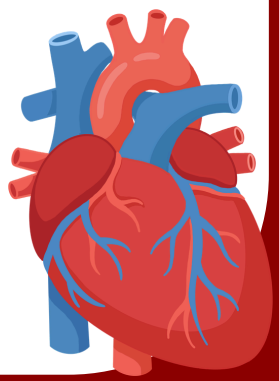
43. Case about Patient have temporal headache, Whats the vascular disease?
A) kaposi sarcoma
B) malignant hypertension
C) giant cell arteritis
D) Kawasaki disease

45. Reduced ejection fraction in which of the following ?
A) systolic dysfunction
B) diastolic dysfunction
C) Exercise

46. A female has controlled hypertension for months which of the following is LEAST likely to occur?
A) Mönckeberg's arteriosclerosis
B) atherosclerosis
C) Myocardial infarction

48. Case: a patient with heart murmur, what is the least expected finding?
A) aschoff bodies (Testbank answer)
B) fibrinous pericarditis (AI answer)
C) fused valves

42	43	45	46
A	C	A	A



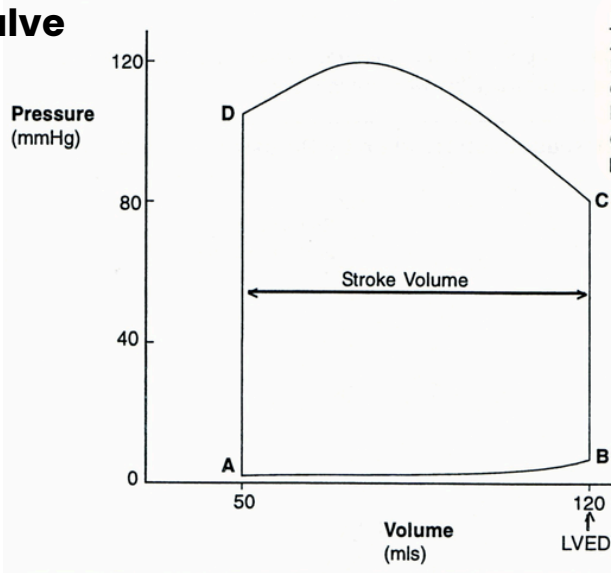
MED21 midterm

1) On cardiac examination, the patient's rhythm was normal with S4 at the apex. What characterizes the S4 sound?

A)it's caused by atrial relaxation during ventricular ejection
B)it can be heard in patients with left ventricular hypertrophy
C)it can heard with the stethoscope diaphragm
D)it has a high frequency, usually 600cycles/ sec

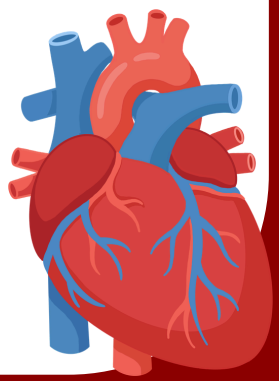
14) At which point does the mitral valve closure occur?

a- B
b- A
c- D
d- C



16) Patient came to the hospital and the doctor diagnosed with heart failure but the doctor didn't request ECG because?

a- ECG is not diagnostic of heart failure
b- ECG expensive
c- ECG aggregates heart failure symptoms
d- ECG invasive



1	14	16
B	A	A

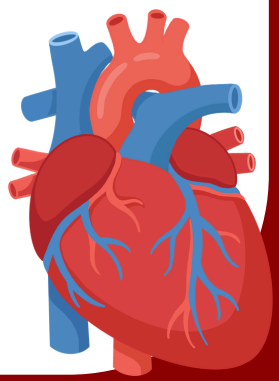
MED21 midterm

19)Which of the following primitive structure develops into left auricle ?
a- primitive atrium
b- bulbus cordis
c- pulmonary veins
d- sinus venosus

24) 70 years old man have pulsating palpable abdominal mass he has hypertension and controlled by antihypertensive drug and he is heavy smoker for 40 years what is the pathogenesis of his condition?
a- Weakening of the blood vessels due to atherosclerosis
b- Weakening of the blood vessels do to cystic media
c- Fibrosis of blood vessels
d- Destruction of media

27) What wave in the ECG represents ventricular repolarization?
a- Q wave
b- R wave
c- T wave
d- P wave

31) defect in the heart that cause overrides both ventricles and receives blood from both sides what is the reason?
a- interventricular septal fail to develop
b- Conotruncal septal fail to develop
c- valvular defect
d- open ductus arteriosus



19	24	27	31
A	A	C	B

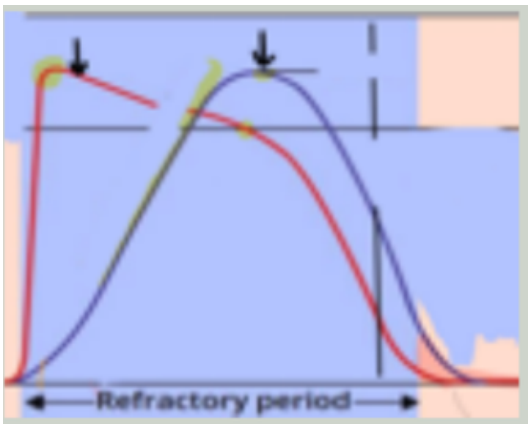
MED21 midterm

34) what structure can cause Atrial septal defects (ASD) through the opening of?

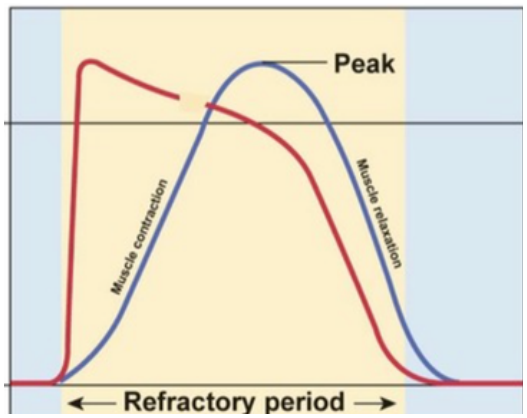
a- foramen ovale
b- patent ductus venosus
c- patent ductus arteriosus

38) what is the correct about blue line?

a- end of depolarization
b- line between systole and diastole
c- line between ARP and RRP
d- both of systole and diastole



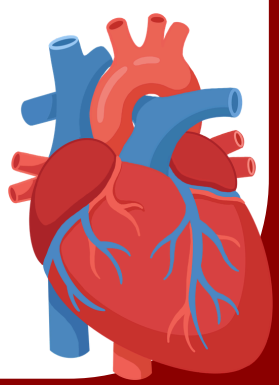
(Testbank photo)



(A clearer photo I found)

Answer : C (the doctor told us the right answer during exam without explaining)

34	38
A	C



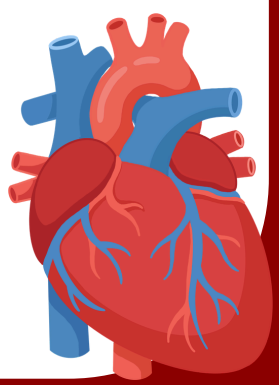
Final CVS Exam MED 24

16 - When is combination antihypertensive therapy typically initiated as first-line treatment?
- A) Stage 1 Hypertension with BP > 130/80 mmHg
- B) Stage 2 Hypertension with BP > 160/100 mmHg
- C) Isolated systolic hypertension in the elderly
- D) Pre-hypertension with lifestyle modification

19-A painful subcutaneous nodule that is relieved by surgical excision is most characteristic of:
- **A)** Glomus tumor (Glomus body)
- **B)** Lipoma
- **C)** Kaposi sarcoma
- **D)** Bacillary angiomatosis

20-Secondary hypertension associated with hypokalemia is most commonly caused by:
- **A)** Primary hyperaldosteronism (Conn's syndrome)
- **B)** Renal artery stenosis
- **C)** Hypothyroidism
- **D)** Pheochromocytoma

21-Embryologically, the left atrial auricle (left auricle) is derived from which structure?
- **A)** Sinus venosus
- **B)** Primitive atrium
- **C)** Bulbus cordis
- **D)** Truncus arteriosus



16	19	20	21
B	A	A	B

Physiology common protected

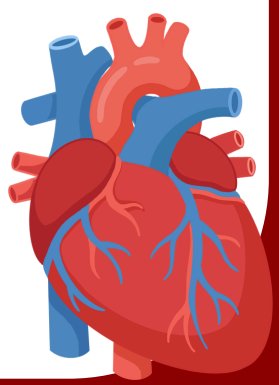
Q1) In a person with heart rate 70 bpm,the duration of cardiac cycle is:
A- 0.85 Sec
B- 0.75 Sec
C- 0.95 Sec
D- 0.65

Q2) If the cardiac cycle is 0.8 sec,the heart rate is expected to be :
A- 70 bpm
B- 80 bpm
C- 90 bpm
D- 100 bpm

Q4) During the isovolumetric contraction phase:
A- The ventricular volume increases
B- The ventricular pressure decreases
C- All valves are closed
D- The second heart sound is heard

Q5) During ejection phase:
A- The AV valves are open
B- The ventricular pressure is lower than the Aortic pressure
C- The ventricular volume is constant
D-The Aortic pressure incre

1	2	4	5
A	A	C	C



Physiology common protected

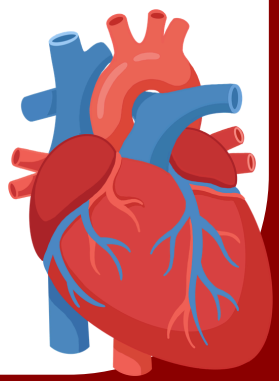
Q6) During the atrial systole:
A- Aortic pressure is increased
B- Ventricular volume decreases
C- Second heart sound is heard
D- Atrial pressure increases

Q7) Aortic pressure increases during:
A- Atrial systole
B- !so-volumetric contraction
C- Rapid ejection
D-Rapid filling phase

Q8) Ventricular pressure shows highest pressure during:
A- Atrial systole
B- !so-volumetric contraction
C- Maximum ejection
D- Reduced ejection

Q9) First heart sound is caused by:
A- Closure of AV valves
B- Opening of AV valves
C- Closure of semilunar valves
D- Opening of semilunar valves

Q10) Second heart sound is caused by:
A- Closure of AV valves
B- Opening of AV valves
C-Closure of semilunar valves
D- Opening of semilunar valves



6	7	8	9	10
D	C	C	A	C

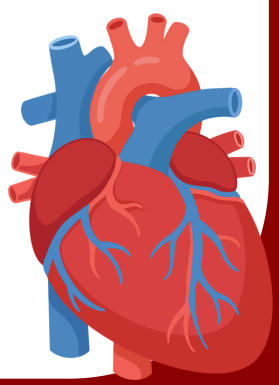
Physiology common protected

Q24) In atrial extra systole:
A- The P wave is absent
B- The premature beat is stronger than the normal one
C- There is rhythm shift
D- There is no depolarization of the S.
E- There is a compensatory pause

Q48) The minimum pressure in the left ventricle during the cardiac cycle is:
A-80 mmHg
B- 25 mmHg
C- 10 mmHg
D- 1mmHg

Q49) Unlike the left ventricle, the pressure of the Aorta during diastole doesn't decrease to 0 mmHg due to the:
A- Recoil tendency of the Aorta (Elasticity)
B- Closure of the Aortic valve
C- Venous return
D- Sympathetic tone

Q59) The ventricles are depolarization during which of the following ECG waves:
A- QRS Complex
B- T wave
C- P-R interval
D- P wave
E- U wave



24	48	49	59
E	D	A	A

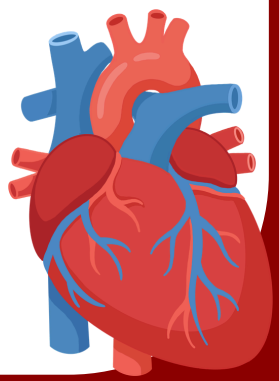
Physiology common protected

Q67) Regarding cardiac cycle:
A. It lasts about .5 seconds w hen the heart rate is 70 beats/minute
B. Diastole shortens at fast heart rates
C. Coronary f low is increased during isometric contraction phase
D. Ventricular

Q74) The important function of Purkinje system in ventricles is to?
a. Delay impulses
b. Select impulses
c. Prevent tetanus in cardiac muscle
d. speed up conduction of impulses

Q77) The diacritic notch on the arterial pulse curve is caused by closure of which of the following valves:
a. Mitral
b. Tric uspid
c. Aortic
d. Pulmonary

Q78) Regarding the heart sound
a. First heart sound results from mitral and tricuspid closure
b. Fourth heart sound may be ausdible normally
c. Second heart sound occur duringthe isometric contraction phase
d. The first heart sound is normally spitted during inspiration



67	74	77	78
B	D	C	A

Physiology common protected

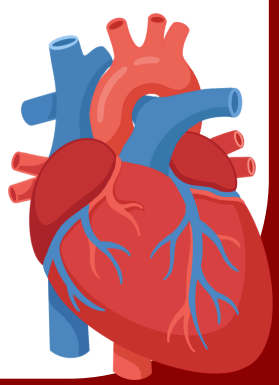
Q85)In isometric relaxation phase of the cardiac cycle
a. No heart sound are heard
b. The atrial pressure is unchanged
c. The ventricul ar pressure is decreased
d. The ventricul ar volume is increased

Q86) The first heart sound corresponds in time which
a. Closure of the AV valves
b. A rise in atrial pressure
c. A decrease in ventricular pressure
d. The A wave in central venous pressure

Q87) During which phase of the cardiac cycle the aortic pressure is lowest ?
a. Atrial systole
b. Isometric phase of contraction
c. Isometri c phase of relaxation
d. Reduced ejection phase

Q88) With reference to the mechanical events in the cardiac cycle?
a. Left ventricle ejects more blood than the right ventricle
b. Mitral valve opens at the maximum filing phase
c. Venous return increases atrial pressure causing the c wave
d. During diastole the left ventricular pressure isabout 70 mmHg

85	86	87	88
C	A	B	B



Physiology common protected

Q99) During the isometric ventricular contraction of the cardiac cycle

a. The pressure inthe aorta rises
b. The atrial pressure is decreased
c. All valves of the heart are closed
d. The ventricul ar volume is increased
e. The ventricul ar pressure is decreased

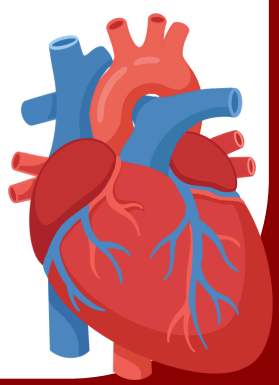
Q102) Which of the following statements regarding ventricular fillingare correct?

a. Normally atrial contraction is responsib le for 50% of ventricular filling at resting heart rate
b. Inflow to the left ventricle is most rapid immediately after the opening of the mitral valve
c. Expiration augment filling of the right ventricle
d. Early rapid ventricular filling is responsible for the c wave of the jugular venous pulse

Q104) During isometric phases of the cardiac cycle

a. All the valves of the heart are opened
b. There is no change inthe tension of the ventricular musxle c. There is no energy expenditure
d. There is a parallel change in both the ventricular & aortic pressures
e. There is no external work do

99	102	104	#
C	B	E	



Physiology common protected

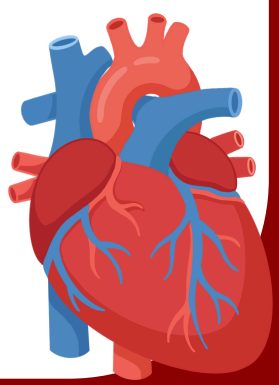
Q105) The first heart sound ?
a. Occurs during the protodiastolic phase
b. Marks the start of diastole
c. Is due to opening of the semilunar valves
d. Non of the above

Q106) The C wave of the JVP occurs ?
a. During the protodiastolic phase of the cardiac cycle
b. At the onset of expiration
c. During the isometri c contraction phase of the cardiac cycle
d. Just before the atrial systole
e. Just after the diastolic notch of the aortic pressure curve

Q114) In which of the following phases in the cardiac cycle all the valves are closed:
a. Atrial systole
b. Isometri c phase of contraction
c. Maximum ejection phase
d. Reduced filling phase

Q115) The "a" wave in the jugular pulse curve is caused by
a. Contraction of the atrium
b. Contraction of the ventricle
c. Closure of the A-V valves
d. Opening of the aortic valve

105	106	114	115
D	C	B	A



Physiology common protected

Q128) All the following occur during isometric contraction phase EXCEPT:
a) Increased intraventricular pressure.
b) Closure of all valves.
c) Atrial pressure becomes higher than ventricular pressure.
d) Production of first heart sound.

129) The period between semilunar valves closure and AV valves opening is known as:
a) Isometric contraction phase.
b) Rapid ejection phase.
c) Isometric relaxation phase.
d) Rapid

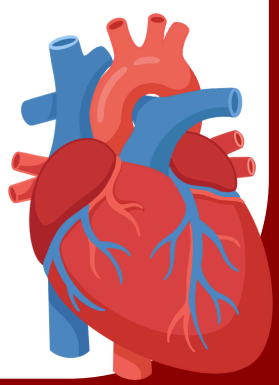
Q139) in the ECG, the diagnosis of heart block is confirmed if:
a) The P-R interval is prolonged.

Q140) Ventricular extra systole is followed by:
a) A Compensatory pause

Q141) Tall T-wave in ECG are suggested of:
a) Hyperkalemia.

Q142) Analysis of ECG can identify:
a) Ischemic changes in the heart

128	129	139	140	141
C	C	A	A	A



Physiology common protected

Q148) in the ECG, the diagnosis of heart block is confirmed if:
a)The P-R interval is prolonged.

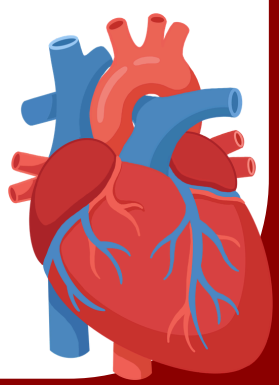
Q149) Ventricular extra systole is followed by:
a) A Compensatory pause

Q150) Tall T-wave in ECG are suggested of:
a) Hyperkalemia.

Q151) Analysis of ECG can identify:
a) Ischemic changes in the heart

Q160) Which of the following events is represented by the ECG?
1. depolarization of SA node
2. Depolarization of AV node
3. atrial repolarization
4. atrial depolarizatio

148	149	150	151	160
A	A	A	A	4



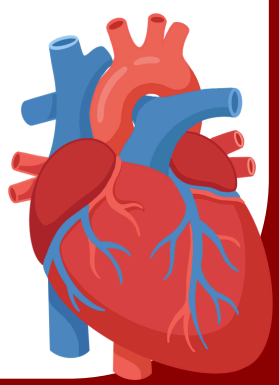
Physiology common protected

Q168) Durig the isovolumetric contraction phase:
a) The contraction of the ventricles is isotonic.
b) The right vent. pressure is increased more than LV.
c) There is neither filling nor ejection.
d) There is external work done.
e) Ventricular volume is decreased.

Q169) During the isovolumetric relaxation phase:
a) All valves are opened.
b) First heart sound is heard.
c) Ventricular volume is increased
d) Ventricular pressure is decreased

Q175)Which of the following events is represented by the ECG?
a. Depolarization of SA node
b. Depolarization of AV node
c. Atrial depolarization
d. Atrial repolarization

168	169	175	#
C	D	C	



CVS M [1]

1. Which of the following is a component of the DASH diet?

A. Low potassium intake
B. Low sodium intake
C. Decreased fruit and vegetable intake
D. High-fat foods

2. A 40-year-old heavy smoker with long-standing hypertension (180/100 mmHg), chronic cough and difficulty breathing requires an antihypertensive. Which option is most appropriate?

A. Amlodipine
B. Losartan

3. Which is a modifiable factor in primary hypertension?

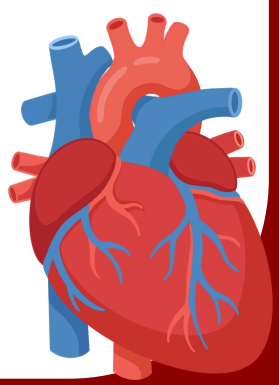
A. Diet and physical exercise
B. Age
C. Family history
D. Ethnicity

4. Blood regurgitates backward into an atrium. Damage to which structure may be responsible?

A. Aortic vestibule
B. Musculi pectinati
C. Chordae tendineae
D. Fossa ovalis

5. A 45-year-old patient has multiple red-blue vascular nodules on the face, containing dilated blood-filled vessels and dystrophic calcification. What is the most likely diagnosis?

A. Capillary hemangioma
B. Cavernous hemangioma
C. Glomus tumor
D. Angiosarcoma



1	2	3	4	5
B	A	A	C	B

CVS M [1]

6. Which vasculitis affects the aorta and its major branches?

A. Takayasu arteritis
B. Polyarteritis nodosa
C. Kawasaki disease
D. Microscopic polyangiitis

7. Which is included in the pathogenesis of primary hypertension?

A. Chronic renal disease
B. Neurogenic disease
C. Adrenal tumor
D. Defective sodium excretion

8. What stimulates the release of atrial natriuretic peptide (ANP)?

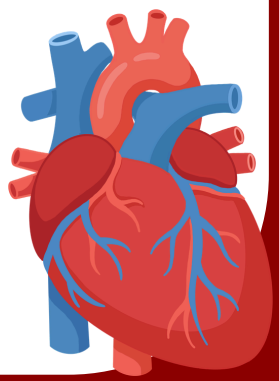
A. Decreased blood pressure
B. Increased stretching of atrial walls
C. Low blood volume
D. Low atrial pressure

9. Which factor contraindicates a renin–angiotensin–aldosterone system–blocking antihypertensive drug?

A. Hypokalemia
B. Smoking
C. Pregnancy
D. Hyperlipidemia

10. A patient reports chronic visual symptoms and has retinal exudates. Which assessment is appropriate?

A. Check vitamin D level
B. Check blood pressure



6	7	8	9	10
A	D	B	C	B

CVS M [1]

11. An athlete has increased inhibitory vagal tone affecting the SA node action potential. Which phase becomes prolonged?

A. Phase 0
B. Phase 3
C. Phase 4
D. Rapid depolarization phase

12. Which is a function of vascular endothelium?

A. Bone formation
B. Blood clot formation
C. Muscle contraction

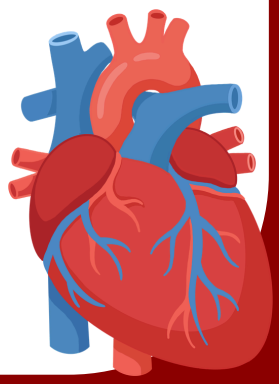
13. Which can cause left ventricular hypertrophy?

A. Increased preload due to [unfinished]
B. Systemic hypertension

14. Which antihypertensive drug is given by IV infusion in a hypertensive crisis?

A. Sodium nitroprusside
[Other options were not provided.]

11	12	13	14
C	B	B	A

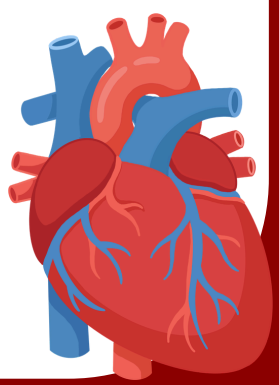


File Name?

1-Maximum left ventricular volume occurs during:
A. Isovolumetric contraction phase
B. Isovolumetric relaxation phase
C. Ventricular ejection phase
D. Rapid ventricular filling phase

2-Maximum coronary blood flow occurs during:
A. Isovolumetric contraction phase
B. Isovolumetric relaxation phase
C. Ventricular ejection phase
D. Atrial systole

1	2	#	#
A	B		

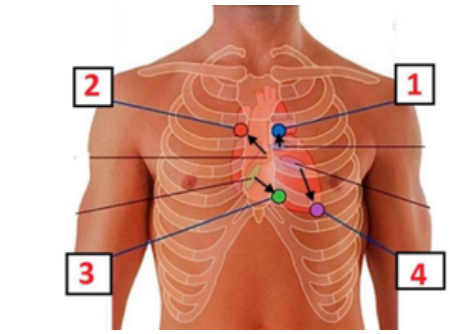


Cvs ospe male afternoon med19_protected

9- auscultatory method > estimate diastolic and systolic

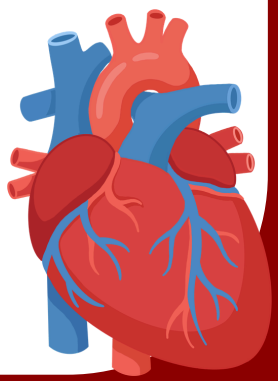
10- A man his pressure 180/90 ,give the pulse and mean pressure:
A) 40mmHg - - - 93mmHg
B) 60mmHg - - - 110mmHg
C) 40mmHg - - - 110mmHg
D) 90mmHg - - - 120mmHg

12- second heart sound is heard best at:
A. 1
B. 2
C. 3
D. 4



14- The blue point on this picture represents which chest lead? **(no picture was given)**
A. avR
B. V5
C. V4
D. II

Q- A 33 year old patient with obstructive heart disease in the ER complaining of fever, chest pain, fatigue. After a few hours the patient died. what was the cause of this patient's death according to the figure shown in the picture. **(no picture was given)**
A. septic
B. sterile



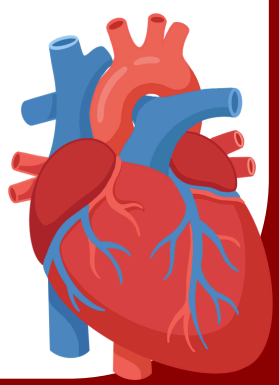
9	10	12	14	Q
-	D	B	C	A

Cvs ospe male afternoon med19_protected

23- Which drug that have an hypertensive effect which abolished with phentolamine?
A- adrenaline
B- noradrenaline

24-Girl 16 years old did take drug that caused hypertension and Abolished by Saralasin , what is the drug?
A-Angiotensin II
B-Phentolamine
C-Hexamethonium
D-Adrenaline

23	24
B	A



CVS Midterm

9- A newborn has a defective coronary sinus, and the cause was mostly congenital, which part of the primitive heart tube is the most likely cause of the defect ?

A)sinus venosus
B)Primitive atrium
C) Primitive ventricle
D) Bulbus cordis

10- From which primitive structure the left auricle arises form?

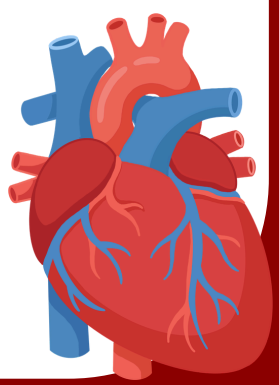
A)Pulmonary veins
B)primitive atrium
C)Bulbus cordis
D)primitive ventricle

11- A defect in the ostium secundum is a defect in which of the following?

A)Upper part of septum primum
B)Upper part of septum secundum
C)Lower part of septum primum
D)Lower part of septum secundum

13- Pulmonary stenosis, VSD, overriding aorta and ventricular hypertrophy presentation is characteristic of which of the following ?

A)tetralogy of fallot
B)Transposition of great arteries
C)Atrial septal defect



9	10	11	13
A	B	A	A

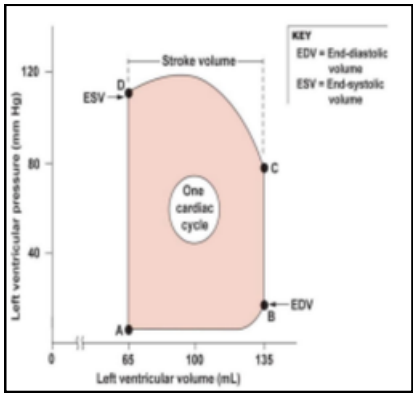
CVS Midterm

21- Male diabetic patient has peripheral vessel disease as he has hypertension.What is contraindicated in this case ?

A)Beta blockers
B)Diuretics
C)Calcium-channel blocker
D)Prazosin

24- Which line represents isovolumetric relaxation phase in the pressure-volume loop of the left ventricle?

A)D-A line
B)B-C line
C)C-D line
D)A-B line



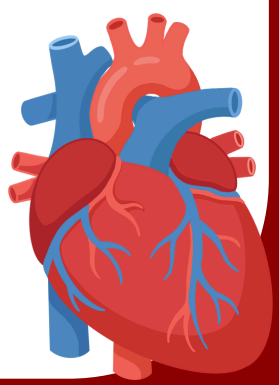
28- ventricular repolarization in ECG is represented by:

A)QRS complex
B)P wave
C)T wave
D)U wave

29- A patient with hypertension and diabetes after several weeks, he became less aware of hypoglycaemia. Which drug is the most likely used ?

A)Atenolol
B)HYdralazine
C)Prazosin
D)Diuretics

21	24	28	29
A	A	C	A



CVS Midterm

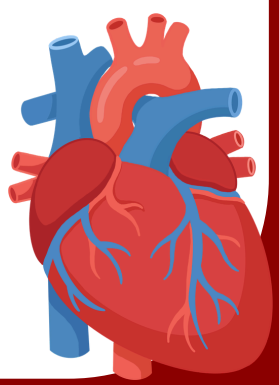
37- increasing radius by 2 fold, decreases the peripheral resistance by what ?

A)32 fold
B)16 fold
C)64 fold
D) 8 fold

38- A patient with hyperthyroidism, which drug mechanism is given to manage cardiovascular problem?

A)increase heart rate
B)Increase ABP
C) Decrease heart rate
D) Decrease ABP

37	38
B	C



Vit.CVS OSPE_protected

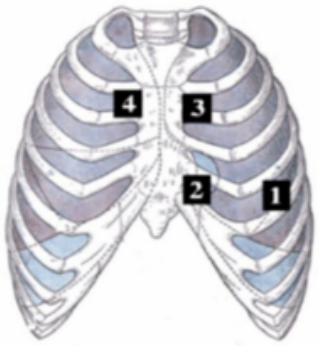
Station 2:

(in question 1 they asked about fossa ovalis)

2- Its embryonic origin is:
A) Primitive atrium
B) Septum primum
C) Septum secondum
D) Sinus venosus

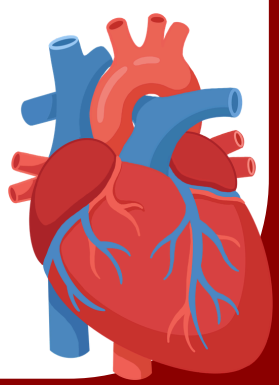
Station 5:

1- Name the auscultatory area numbered 3 in the above photo
A) Mitral area
B) Aortic area
C) Tricuspid area
D) Pulmonary area



2- The cause of the first heart sound is:
A) The closure of semilunar valves
B) The closure of atrioventricular valves
C) Rush of blood from atria to ventricle during rapid filling phase
D) Atrial systole with rapid flow of blood from atria to ventricle

2.2	5.1	5.2
B	D	B



Vit.CVS OSPE_protected

Station 6:

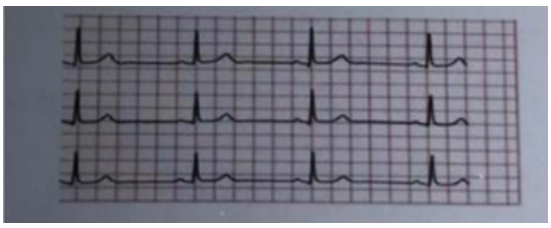
1- The P-R interval of the above ECG tracing equals:

A) 0.06 – 0.10 seconds
B) 0.12 – 0.16 seconds
C) 0.18 – 0.22 seconds
D) 0.24 – 0.8 seconds



2- The heart rate of the above ECG tracing equals:

A) 50 bpm
B) 60 bpm
C) 75 bpm
D) 100 bpm



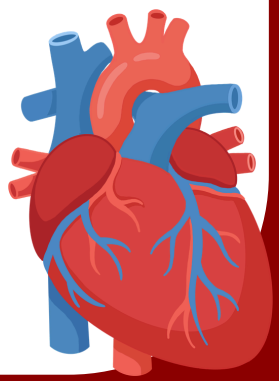
Station 8:

1- A medical student is involved in a summer research project involving the use of an unknown agent. The infusion of the unknown agent into an animal model produced hypertensive effect that was abolished by saralasin. What is the mechanism of action of the unknown drug?

A) Angiotensin II receptor agonist
B) Vasopressin receptor agonist
C) Alpha adrenoreceptor agonist
D) Nicotinic receptor agonist

2- If injected nicotine intravenously into an anesthetized cat, which of the following effect would be expected?

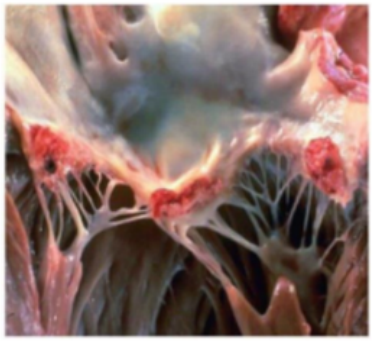
A) Hypotensive effect that is reversed by propranolol
B) Hypertensive effect that is reversed by phentolamine
C) Hypertensive effect that is abolished by phentolamine
D) Hypertensive effect that is antagonized by phentolamine



6.1	6.2	8.1	8.2
B	A	A	B

Vit.CVS OSPE_protected

Station 9:



1- Which of the following describes the lesion?
A) Irregular bulky destructive valve
B) Warty non destructive valve with commisural fusion
C) Enlarged, rudent, thick, rubbery

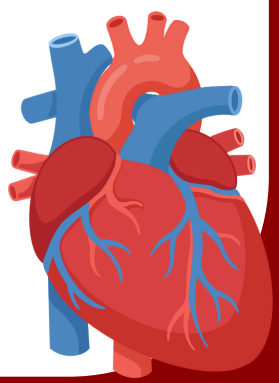
2- What is the cause of the morphology?
A) Chronic rheumatic fever
B) Acute rheumatic fever
C) Acute infective endocarditis
D) Connective tissue disease

Station 11:



1- The most likely diagnosis of this lesion is:
A) Aortic aneurism
B) Dissection of the aorta
C) Infective endocarditis with aortic stenosis
D) Cardiac myxoma

2- Which of the following is considered a complication of this lesion?
A) Left ventricular hypertrophy
B) Aneurysm of abdominal aorta
C) Brown atrophy of the heart
D) Right ventricular hypertrophy



9.1	9.2	11.1	11.2
A	C	C	A

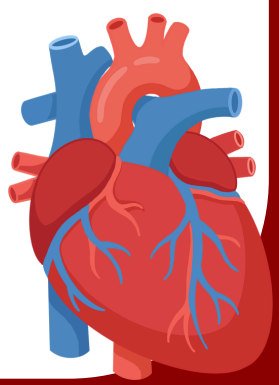
Cvs final afternoon med19_protected

11- QRS complex of ECG occur immediately before which phase ?
A- Isometric contraction
B- Isometric relaxation
C- Atrial systole
D- Maximum filling phase

12- Lead 2 is :
A/ +ve in Right arm and -ve in left arm
B/ +ve in Right arm and -ve in left leg
C/ +ve in Left leg and -ve in Right arm
D/ +ve in Left leg and -ve in Left arm

14- What is the cause of the ecg in the picture above
(no picture was given)
A. Hyperkalemia
B. Hypokalemia
C. Hypercalcemia
D. Hypocalcemia

11	12	14
A	C	A



File - CVS FINAL

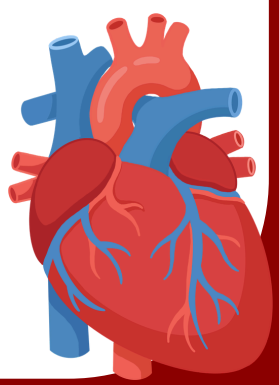
3- Which of the following is an external feature of the right atrium?
A. rough auricle
B. Sulcus terminalis
C. Pectinate muscles
D. Cristea terminalis

4- The MI most commonly affects which portion of the left ventricle (outflow area)?
A. Infundulum
B. Vestibule
C. Mitral valve
D. Papillary muscle

5-Which of the following is part of the anatomy of the atrioventricular valves?
A. Same number of leaflets
B. Nodule
C. Sinus
D. Papillary muscle

12- Frank-Starling law states?
A. Increasing end-systolic volume increases contractility
B. Increasing end-diastolic volume increases contractility
C. Decreasing venous return, increases cardiac output

3	4	5	12
B	B	D	B



File - CVS FINAL

13- A patient lost 300ml of blood after a hemorrhage, what is the most likely physiological response?

A. Decreased heart rate
B. Increased arterial blood pressure
C. Decreased central venous pressure
D. Increased cardiac output

18- Someone has rib scoliosis; from your understanding, how will this affect his hemodynamic mechanism?

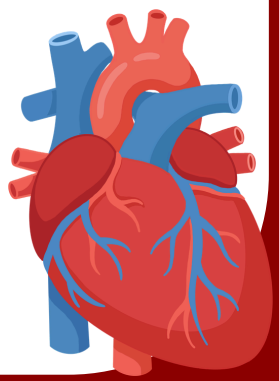
A. Thoracic has no effect on
B. Decreased venous return due to decreased thoracic widening in inspiration
C. Increased venous return due to increased thoracic widening in inspiration

21- A patient has the ventricles sharing one outflow tract, what is the most likely defect?

A. Ventricular septal defect
B. Conotruncal septum not formed
C. Incomplete development of endocardial cushions

22- A patient has a thin outflow part of the right atrium, most likely defect?

A. body & right sinus horn
B. Couns cordis
C. Primitive ventricle



13	18	21	22
D	B	A	A

File - CVS FINAL

25- vestibule of left ventricle is formed of ?

A. Sinus venosus
B. Primitive atrium
C. Primitive ventricle
D. Bulbus cordis

26- Failure of the conotruncal ridge twisting?

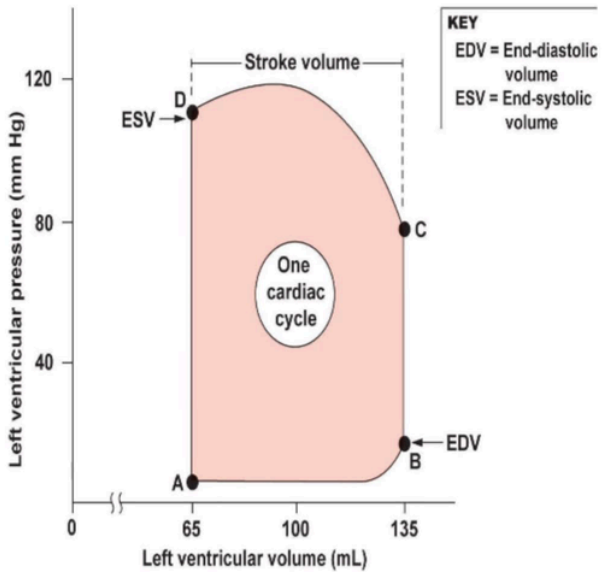
A. Transposition of the great vessels
B. VSD
C. ASD

27- The phase of the cardiac cycle between mitral valve closure and aortic valve opening?

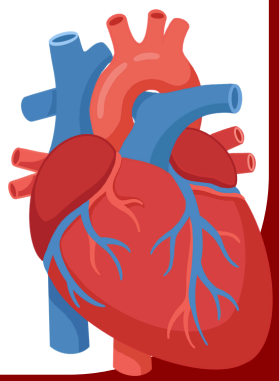
A. Isovolumetric contraction phase
BAtrial systole
C. Ventricle systole
D. Isovolumetric relaxation phase

28- Which point represents mitral valve closure?

A
B
C
D



25	26	27	28
D	A	A	B



File - CVS FINAL

29- A patient with low arterial blood pressure, what is the reflex?
A. Decreased carotid sinus firing rate
B. Decrease heart rate
C. Decrease contractility
D. Increase vagal tone

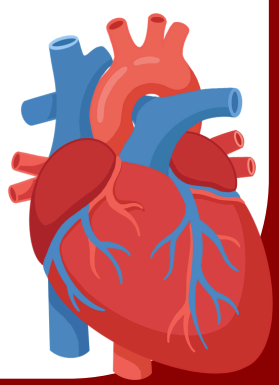
30- A person with a headache has hypertension. What compensatory response is initiated ?
A. Bezold Jarish reflex
B. Bainbridge reflex
C. Baroreceptor
D. Chemoreceptors

32- A patient's biopsy of the small blood vessels revealed pink homogenous hyaline tissue, most likely cause?
A. Malignant hypertension
B. Benign hypertension

3r- The following shows what complication? PICTURE
A. Malignancy of the blood vessel
B. Ulcerated atheroma
C. Fatty streak



29	30	32	3r
A	C	B	B



File - CVS FINAL

38-A patient came with excruciating chest pain to the ER, he died, and their biopsy showed the descending thoracic aorta of normal caliber had ruptured, found a blood filled space in the middle layer of the blood vessel, most likely cause of death?
A. Aortic dissection
B. Abdominal aorta aneurysm
C. Atherosclerosis

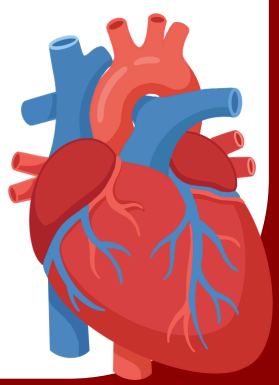
What is a predisposing factor to heart failure?
A. Acute MI
B. Exercise

40. Patient has an abdominal palpable pulse, most likely cause?
A. Abdominal aorta aneurysm

41.A patient came to the ER with chest pain, they have a history of uncontrolled hypertension, they died, most likely cause of death?
A. Fibrosis of heart tissue
B. Reversible injury
C. Necrosis

43. **Patient came in with acute MI and died, the most likely cause of death?**
A. Arrhythmia
B. Ventricular hypertrophy (not sure)

38	#	40	41	43
A	A	A	C	A



File - CVS FINAL

44) Which of the following is stage 2 hypertension?
A) 180/100
B) 135/70

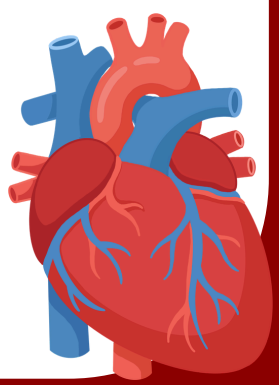
45) A diabetic patient came in with masked hypoglycemia symptoms after starting a drug, this drug is most likely?
A) Atenolol

46) A diabetic patient came in with chest pain, they also have intermitent claudication (peripheral vascular disease), which drug is contraindicated for this patient?
A. Captopril
B. A beta blocker

47) ACEI's and ARBs share similar adverse reactions, but what do only ACEI's cause?
A. Dry cough
B. Hypotension

48) A patient was given an ACEI and got arrhythmia problems, what it the most likely ion imbalance cause (not 100% sure of how right the question is here????) a tall peaked t-wave was visible on ECG
A. hyperkalemia
B. hypokalemia
C. hypercalcemia

44	45	46	47	48
A	A	B	A	A



File - CVS FINAL

50) A patient was given two doses of adenosine but still the ___ remains, what is the best drug to give?
A. Amiodarone
B. beta blocker or verapamil [was there verampil or beta blockers in the options? i forgot what the problem was but the edited sais: if adenosine doesnt work to treat acute paroxysmal SV tachycardia then we give them a beta blocker or verampil]

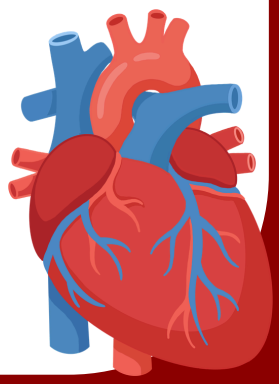
53) A patient is on an anti-anginal drug but took a holiday and forgot his medication, this would be disastrous if the drug was most likely?
A. methyl dopa
B. clonidine
C. pindolol

55) Mechanism of action of aspirin?
A) inhibits COX

61.Which drug will decrease the sympathetic response and renin secretion?
A. ACE inhibitor
B. beta blocker

62) Which is a primary cause of hypertension?
A. Chronic renal failure
B. Defect in sodium excretion
C. neurogenic cause
D. Kidney tumor

50	53	55	61	62
B	B	A	B	B



File - CVS FINAL

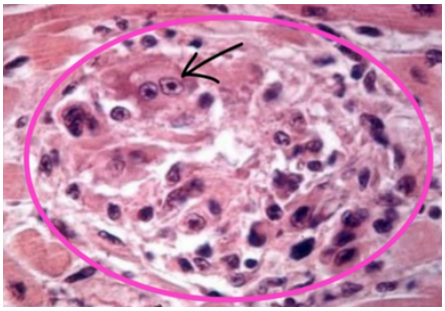
63) Decrease sodium retention helps with hypotension how?
A) Decreasing body fluid

64) In a healthy heart, the QRS waves correspond to:
A. Ventricular repolarization
B. Ventricular depolarization
C. Atrial repolarization
D. Atrial depolarization

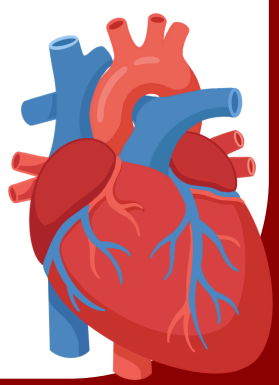
65) In first degree heart block what can be expected?
A) PR interval is less that 0.12
B) PR interval is prolonged
C) P wave is sometimes followed by QRS
D) P wave is not followed by QRS

66.A patient has an elevated ST segment and inverted T wave, diagnosis?
A) acute Mi(ischemic heart disease)

67.Patient presented with migratory polyarthritis and fever, their biopsy showed the following, what is it? PICTURE
A) Anitschkow cells characteristic in acute rheumatic heart disease
B) Plump plasma cells characteristic in rheumatic heart disease
C) Foreign body Giant cell characteristic in acute rheumatic heart disease
D) Aschoff cells characteristic in mitral valve prolapse



63	64	65	66	67
A	B	A	A	C



File - CVS FINAL



68) Which valve is most likely affected by this?
A. Aortic valve
B. Mitral valve
C. Pulmonary valve
D. Tricuspid valve

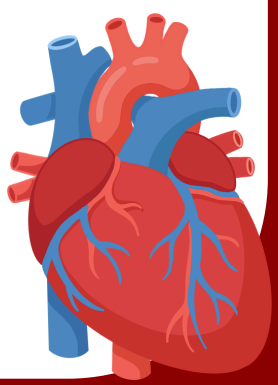
69) Which bacteria is most likely to cause infective endocarditis in IV drug users?
A. Staph. Auerus
B. B. hemolytic bacteria

70) A patient with SLE has vegetations on his valves, what is the cause?
A) Immune complex deposition

71) Which of the following is likely to spontaneously regress?
A) Strawberry hemangioma

72) A patient with von Hippel-Lindau disease has which of the following?
A. Cavernous Hemangioma
B. Glomus Tumor
C. Angiosarcoma

73) Kaposi Sarcoma is caused by which of the following?
A. Human herpesvirus-8 (HHV-8)
B. Human papilloma virus



68	69	70	71	72	73
B	A	A	A	A	A

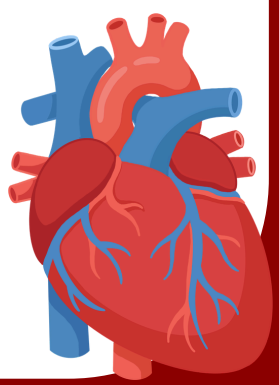
File - CVS FINAL

74) A female patient had gotten radiation and a mastectomy for her breast cancer, she is nowpresenting with ___? What is her current condition?
A. Epithelioid Hemangioendothelioma
B. Angiosarcoma

75) What is true about coronary circulation?
A. The heart gets a rich blood supply from the right and left coronary arteries that arise from the coronary sinus
B. During ventricular systole the heart presses the arteries causing them to obstruct
C. Ventricles only fill during diastole

77)A woman was shopping at a mall, suddenly she had chest pain with vasospasm. Immediately she used nitroglycerin and it relieved her symptoms. What is the mechanism of action of nitrates to alleviate chest pain? (not exact wording)
A. decrease pulmonary arterial pressure
B. Decrease 02 demand

80)Case about a patient of heat intolerance, weight loss, hyp_tension, ... what do we test for?
A. Thyroid stimulating hormone
B. Concentrations of (sth with Mine)
C. serum Cortisol
D. fasting insulin (or sth like that)



74	75	77	80
B	B	B	A

File - CVS FINAL

82)Case: A 40 years old heavy smoker came to the hospital with long term hypertension 180/100, he has a cough and difficulty breathing, which medication is best to administer to control his BP?

A. Amlodipine

B. Losartan

84) How does low sodium help with treating hypertension?

A. decrease fluid buildup

B. allow vessels to relax

86) A patient with MI showed a destruction of the cardiac muscle fibers. Which of the following cell junctions at intercalated disks and serve for keeping the integrity of the cardiac muscle during contraction are expected to be damaged in this condition?

A) fascia adherens

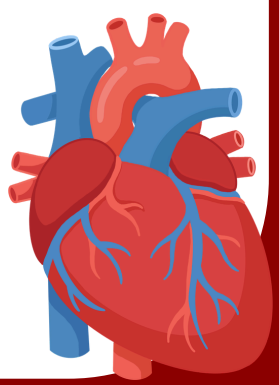
B) zonula occudens

C) macula adherens

D) gap junction

Gra

82	84	86
A & B	B	B



Final CVS [1]

42. Second heart sound produced by closure of which valve? (ECE)

A) Aortic and pulmonary
B) Mitral and Aortic
C) Aortic and tricuspid
D) tricuspid and pulmonary

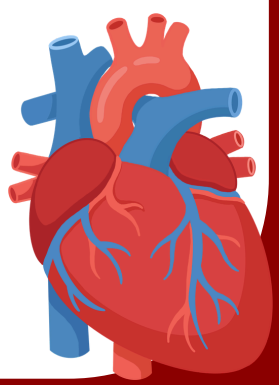
1. **(Book Q.) A 3-year-old child is developing normally. Physical examination reveals a low-pitched cardiac murmur. An echocardiogram shows the presence of an ostium secundum, with a 1-cm defect. Which of the following abnormalities is most likely to be found in this child?**

A) Cyanosis at rest
B) Left-to-right shunt
C) Mural thrombosis
D) Pericardial effusion

6. **Which of the following mediators causes vasodilation?**

A. Nitric Oxide

42	1	6
A	B	A



Final CVS [1]

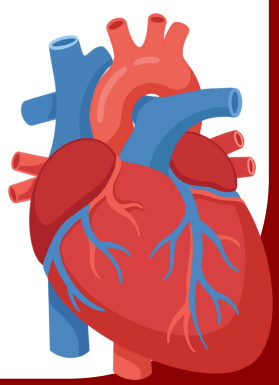
42. Second heart sound produced by closure of which valve? (ECE)
A) Aortic and pulmonary
B) Mitral and Aortic
C) Aortic and tricuspid
D) tricuspid and pulmonary

8. **Which of the following is considered to be the first step in the pathogenesis of atherosclerosis?**
A) Endothelial Dysfunction

9. **A 41-year old diabetic male has damaged vascular endothelium. Advanced glycation end products (AGEs) are present in this patient. Which of the following could result from such findings?**
A) Blood coagulopathy
B) Recurrent infections

10. **In a normal ECG, which of the following coincides with the absolute refractory period of a myocardial cell?**
A) Start of Q to top of T wave

42	8	9	10
A	A	A	A



Final CVS [1]

.segments from leads II, III, and avF. Which of the following is most likely seen in this patient?

A) Infarct in the inferior part of the heart with congestive heart failure

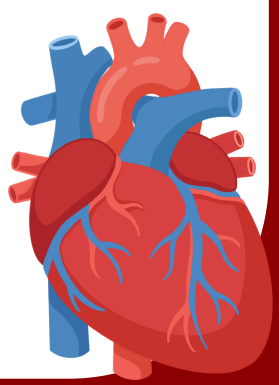
B) Infarct in the right part of the heart with hypotension

C) Infarct in the left part of the heart with bradycardia

D) Infarct in the inferior part of the heart with normal ventricular function

13. **Which of the following heart sounds can be heard as a result of rapid blood filling from the atria into the ventricles (gushing)?**

A) First heart sound



#	13
A	A

Final CVS [1]

17. **Which of the following primitive structures develops into the left auricle?**

A) bulbus cordis
B) primitive atrium
C) pulmonary veins
D) sinus venosus

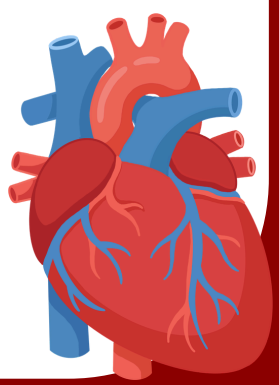
18. **Which structure can cause an atrial septal defect (ASD) through an opening?**

A) Foramen ovale
B) Patent ductus venosus
C) Patent ductus arteriosus

.hypotensive, with clammy and sweaty hands. Which of the following is most likely the type of shock?

A) Cardiogenic
B) Septic
C) Hypovolemic
D) Distributive

17	18	#
B	A	C



Final CVS [1]

22. **Which of the following is contraindicated for initiating a beta blocker for congestive heart failure?**

A) Recent decompensated HF
B) Ventricle ejection fraction < 40
C) Current digoxin use

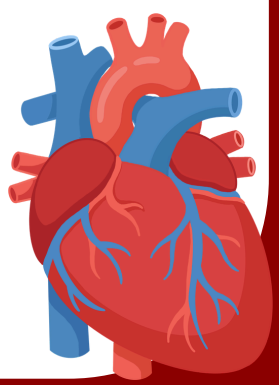
27. **What are giant arteritis cells?**

A) Necrotizing vasculitis,
B) lymphatic infiltration,
C) macrophages
D) Granulomatous inflammations

28) **A woman presents to the clinic with a painful mass under her fingernail. Biopsy of the mass shows modified smooth muscle cells with stroma. What's the diagnosis?**

A) Glomus tumor

22	27	28
A	D	A



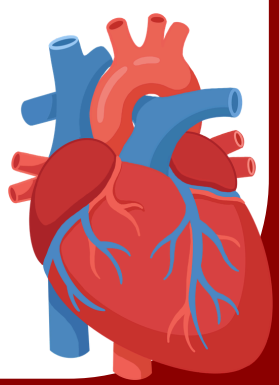
Final CVS [1]

.36. **What is the most common heart condition that presents with an irregularly irregular rhythm? (ECE)**
A) Atrial Flutter
B) Ventricular Tachycardia
C) Atrial Fibrillation
D) Ventricular Fibrillation

37. **Which of the following is the most accurate description of Janeway lesions? (ECE)**
A) Painful nodules in the pulp of fingers
B)Not painful spots on palms and soles of feet

38) **Smoking is an important element of social history. Which of the following is a consequence or complication of smoking? (ECE)**
A)Obesity
B) epilepsy
C)premature coronary artery disease

36	37	38
A	B	C



Final CVS [1]

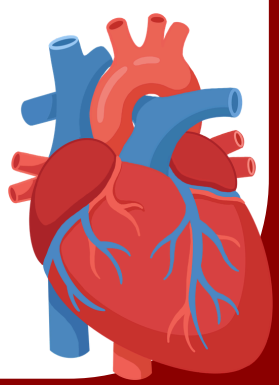
40. **Someone came with dyspnea, a cough that produces frothy sputum with wheezing. And heard with a stethoscope crackles, what is the diagnosis? (ECE)**
A) pulmonary edema
B) asthma
C) pneumonia
D) congestive heart failure

41. **The correct position in a cardiovascular examination? (ECE)**
A) 45 degrees
B) 90 degrees
C) standing position
D) supine

43. **A 25-year-old male patient has a past history of rheumatic fever. His auscultatory findings of the heart showed a loud first heart sound and a mid-diastolic murmur at apex. Which of the following is the most likely diagnosis? (ECE)**
A) Aortic regurgitation

44. **Case with a patient that had splinter hemorrhage. Most likely diagnosis? (ECE)**
A) Infective Endocarditis
B) Heart failure
C) Renal failure

40	41	43	44
A	A	A	A



Final CVS [1]

45. **A 52-year old man develops sudden severe central chest pain after lifting heavy cases while moving to a new house. The pain radiates to the back and both shoulders but not to either arm. His BP is 170/110 mmHg, pulse rate is 92 beats per minute and the ECG is normal. He is distressed and sweaty, but not nauseated. What would you cinder the most likely diagnosis? (ECE)**

A) Myocardial Infarction
B)Aortic Dissection

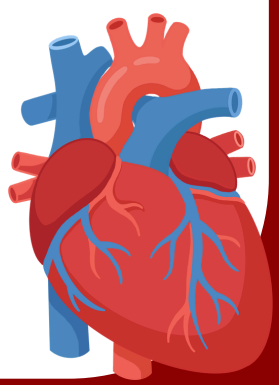
46. **How is the SA node rate affected by sympathetic stimulation?**

A) increase force of ventricular contraction
B) prolonged AV nodal dely
C)aortic blood pressure
D) increase slope of prepotential SA node

47. **Which of the following is directly proportional to the rate of blood flow?**

A) Vessel radius
B)Vessel length
C) Vessel viscosity
D) Hematocrit value

45	46	47
B	D	A



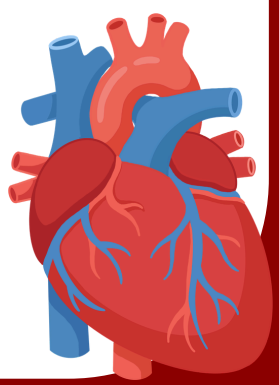
Final CVS [1]

48. **A decrease in the slope of the prepotential SA node can be caused by?**
A) Increased vagal stimulation

49. **What is the function of Atrio-Ventricular Bundle (AVB)?**
A) Conducts impulses from SA node to ventricles
B) Conducts single from the apex to the rest of the ventricle

52. **What primarily determines heart contractility?**
A)HR
B) Preload (venous return)

48	49	52
A	A	B



Cvs midterm afternoon male med19_protected

9-vascular system developed at :
A) 3rd week

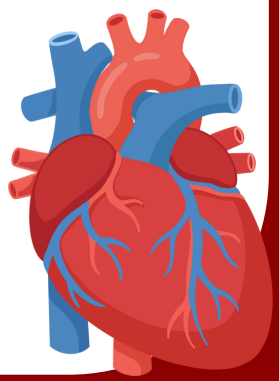
12-ostium secundum is a defect in:
A-upper part of septum primum
B-lower part of septum primum
C-lower part of septum secundum
D-upper part of septum secundum

13- A newly born baby is born cyanotic & having difficulty of breathing , on examination a well-heard diastolic murmur suggested valvular stenosis , the imaging technique used is :
A- Echocardiogram
B- MRI
C- Angiography
D- X-Ray of the skull

21- Between AV valve closure and opening of semilunar valve is the following phase:
A- isometric relaxation
B- rapid ejection
C- isometric contraction
D- reduced ejection

22-the graph shows ventricle pressure-volume relationship; point A is:
(no graph was given)
A- mitral valve closure

23- The lowest ventricular volume occurs during which phase:
A- Isovolmutric ventricular relaxation.
B- Isovolmutric ventricular contraction.
C- Rapid ventricular ejection.
D- Reduced ventricular ejection.



9	12	13	21	22	23
A	A	A	C	A	A

الحمد لله حمداً كثيراً طيباً مباركاً فيه